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COURSE: BCA (AI and DS) (SEC-B) SEM-5

SUBJECT: NLP

ASSESSMENT - 1

Q1) Building Blocks of a language

Language is made up of different basic components called building blocks. The main building blocks are:

- 1) Phonemes - Phonemes are the smallest units of sound in a language. They help distinguish one word from another.
eg: The sounds /c/, /a/, and /t/ from "CAT".
- 2) Morphemes - Morphemes are the smallest meaningful units of a language.
eg: "unhappy" = un + happy.
- 3) Lexemes - A lexeme is a basic vocabulary unit that represents different from of a word.
eg: Run, Runs, Running...
- 4) Syntax - Syntax refers to the rules for arranging words correctly in a sentence.
eg: "The student studies NLP".

5) Semantics - Semantics deals with the meaning of words and sentences.

eg: "The dog is sleeping"

6) Context / Pragmatics - Context helps us understand the intended meaning based on the situation.

eg: "Can you open the window?"

Q2) Challenges Faced While Developing an NLP Project.

While developing an NLP Project, a developer faces many challenges. Three important challenges are:

1. Ambiguity - Ambiguity occurs when a word or sentence has more than one possible.

eg: "The man went to the bank."

Here, bank can mean a financial institution or the side of a river.

2. Creativity - Humans use language creatively through slang, jokes, metaphors, sarcasm and new expressions.

eg: "That exam was a piece of cake."

It does not mean that the exam was made of cake. It means the exam was easy.

3. Common Knowledge - NLP Systems sometimes need background or common knowledge to understand a sentence correctly.
eg: "Rakesh dropped the glass. It broke".

Humans understand that the glass broke because it was dropped.

Q3) Deep Learning - Based Approaches in NLP

Deep Learning (DL) is a machine learning approach that uses neural networks with multiple layers to learn patterns from large amounts of data. It is widely used in NLP.

1. RNN - Recurrent Neural Network (RNN) processes words one after another and uses previous information to understand the current word.

uses: Text generation, sentiment analysis.

2. LSTM - Long Short-Term Memory (LSTM) is an improved version of RNN. It can remember important information for a longer period using special gates.

uses: Translation, text generation & speech.

3. GRU - Gated Recurrent Unit (GRU) is another improved RNN model. It is simpler than LSTM and can process sequences efficiently.

4. CNN - Convolutional Neural Network can identify important local patterns in text.

Uses: Text classification and sentiment analysis

5. Transformer - Transformer uses an attention mechanism to understand relationship between words, even when they are far apart in a sentence.

Eg: BERT, GPT and TS.

Uses: Translation, Summarization, question answering and text generation.

(Q4) Few Shortcomings of Deep Learning models.

Although deep learning models are powerful they also have some limitations.

1. Large Amount of Data - Deep learning models usually require a large amount of training data. Collecting and preparing this data can be difficult and expensive.

2. High computational cost - Training deep learning models requires powerful hardware, such as GPU or TPU. It also requires high memory, time & energy.
3. Lack of Interpretability - Deep learning models are often called black-box models because it can be difficult to understand why the model made a particular prediction or decision.
4. Overfitting - Happens when a model learns the training data too closely but performs poorly on new or unseen data.

for ex: a model may give very accurate results on training data but give incorrect results on test data.

Conclusion

Therefore, The major shortcomings of deep learning models are large data requirements, high computational cost, lack of Interpretability and overfitting.